R18

Code No: 156DK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
BASICS OF SENSORS TECHNOLOGY
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Define potentiometer. Explain about the resistive potentiometers in detail.
b) Briefly explain about the dynamic characteristics of sensors. [8+7]

2.a) What is RTD? Explain the relationship between resistance and temperature for the RTD with temperature resistance curve.
b) What are differential capacitive sensors? Explain the function of a capacitive sensor in a robot end Effectors. [7+8]

3. Explain the working of piezo electric sensors. State their applications. [15]

4.a) Explain about the measurement of force using Strain gauge load cell method.
b) Explain how to measure rotational velocity using stroboscopic methods. [8+7]

5.a) What is Gauge factor? Derive an expression for Gauge factor in terms of poisson's ratio.
b) Explain principle of operation of L.V.D.T. with its characteristics. [8+7]

6.a) With a neat diagram, explain in detail about Oscillating Type Consistency Meter.
b) Explain in detail about differential pressure densitometer. [8+7]

7. Explain different methods for measuring consistency and viscosity. [15]

8. Explain in detail for following interfacing sensors.
a) Force sensors
b) Viscosity sensors
c) Variable frequency device. [5+5+5]

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